

Proof portfolio Math 314

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Proof portfolio part I

Lemma 1. *If n is any integer, then $n^2 + n$ is even.*

Proof. Let $n \in \mathbb{Z}$.

We have $n^2 + n = n(n + 1)$.

Assume n is even, so $\exists k_1 \in \mathbb{Z}$, s.t. $n = 2k_1$.

$$n^2 + n = 2k_1(n + 1)$$

$\exists k_2 = k_1(n + 1) \in \mathbb{Z}$, s.t. $n^2 + n = 2k_2$, so $n^2 + n$ is even if n is even.

Assume n is odd, so $\exists k_3 \in \mathbb{Z}$, s.t. $n = 2k_3 + 1$.

$$\begin{aligned} n + 1 &= 2k_3 + 1 + 1 = 2(k_3 + 1) \\ n^2 + n &= n(n + 1) = n(2k_4) = 2(nk_4) \end{aligned}$$

$\exists k_4 = k_3 + 1 \in \mathbb{Z}$, s.t. $n + 1 = 2k_4$, so $n + 1$ is even.

$\exists k_5 = nk_4 \in \mathbb{Z}$, s.t. $n^2 + n = 2k_5$.

So $n^2 + n$ is even if n is odd.

Overall $n^2 + n$ is even if $n \in \mathbb{Z}$. □

Lemma 2. *Let x and y be integers. Prove that $x^2 + xy$ is odd if and only if x is odd and y is even.*

Proof. (Forward) (Proof by contradiction) Let $x^2 + xy$ is odd.

$$\exists k_1 \in \mathbb{Z}, s.t. x^2 + xy = 2k_1 + 1$$

If x is even: $\exists k_2 \in \mathbb{Z}, s.t. x = 2k_2$

$$x^2 + xy = 4k_2^2 + 2k_2y = 2(2k_2^2 + k_2y)$$

So, $\exists k_3 = 2k_2^2 + k_2y \in \mathbb{Z}, s.t. x^2 + xy = 2k_3$, $x^2 + xy$ is even, contradiction, x is odd.

If y is odd: $\exists k_4 \in \mathbb{Z}, s.t. y = 2k_4 + 1$

$$x^2 + xy = x^2 + (2k_4 + 1)x = x(x + 2k_4 + 1)$$

Since x is odd (Proof above): $\exists k_5 \in \mathbb{Z}, s.t. x = 2k_5 + 1$

$$x(x + 2k_4 + 1) = (2k_5 + 1)(2k_5 + 2k_4 + 2) = 2[(2k_5 + 1)(k_5 + k_4 + 1)]$$

$$\exists k_6 = (2k_5 + 1)(k_5 + k_4 + 1) \in \mathbb{Z}, s.t. x(x + 2k_4 + 1) = 2k_6$$

So $x(x + 2k_4 + 1)$ is even, contradiction.

So $x^2 + xy$ is odd if x is odd and y is even.

(Backward) Let x be odd and y be even.

$$\exists k_7 \in \mathbb{Z}, s.t. x = 2k_7 + 1$$

$$\exists k_8 \in \mathbb{Z}, s.t. y = 2k_8$$

$$x^2 + xy = x(x + y) = (2k_7 + 1)(2k_7 + 1 + 2k_8)$$

$$= 4k_7^2 + 2k_7 + 2k_7 + 1 + 4k_7k_8 + 2k_8$$

$$= 2(2k_7^2 + k_7 + k_7 + 2k_7k_8 + k_8) + 1$$

$$\exists k_9 = 2k_7^2 + 2k_7 + 2k_7k_8 + k_8 \in \mathbb{Z}, s.t. x^2 + xy = 2k_9 + 1$$

So $x^2 + xy$ is odd. □

Lemma 3. If x and y are real numbers with $xy > 0$, then $\frac{x}{y} + \frac{y}{x} \geq 2$.
Determine when equality holds.

Proof. Since $(x - y)^2 \geq 0$,

$$\Rightarrow x^2 - 2xy + y^2 \geq 0$$

$$\Rightarrow x^2 + y^2 \geq 2xy$$

Since $xy > 0$,

$$\frac{x^2 + y^2}{xy} \geq \frac{2xy}{xy}$$

$$\Rightarrow \frac{x}{y} + \frac{y}{x} \geq 2$$

The equality holds when $(x - y)^2 = 0$, $x = y$.

□

Let $(x_n) = (x_1, x_2, x_3, \dots)$ denote a sequence of real numbers. Consider the following definitions

Definition. " (x_n) diverges to ∞ " means: $\forall M > 0, \exists N \in \mathbb{R}$ such that $n > N \Rightarrow x_n > M$

Definition. " (x_n) converges to L " means: $\forall \epsilon > 0, \exists N \in \mathbb{R}$ such that $n > N \Rightarrow |x_n - L| < \epsilon$

Definition. " (x_n) does not to diverge to ∞ " means: $\exists M > 0, \forall N \in \mathbb{R}$ such that $n > N$ and $x_n \leq M$

Definition. " (x_n) does not converge to L " means: $\exists \epsilon > 0, \forall N \in \mathbb{R}$ such that $n > N$ and $|x_n - L| \geq \epsilon$

Definition. " (x_n) does not to converges at all" means: $\forall L \in \mathbb{R}, \exists \epsilon > 0, \forall N \in \mathbb{R}$ such that $n > N$ and $|x_n - L| \geq \epsilon$

In the next two Lemmas, all variables are integers.

Lemma 4. If $a \mid b$ and $c \mid d$ then $ac \mid bd$.

Proof. $\exists k_1 \in \mathbb{Z}$, such that $\frac{b}{a} = k_1$, and

$\exists k_2 \in \mathbb{Z}$, such that $\frac{d}{c} = k_2$.

So $k_1 k_2 = \frac{bd}{ac}$,

and $k_1 k_2 \in \mathbb{Z}$.

So $\exists k_3 = k_1 k_2 \in \mathbb{Z}$, such that $\frac{bd}{ac} = k_3$.

$\Rightarrow ac \mid bd$. □

Lemma 5. *If $a \mid b$ and $a \mid c$ then $a \mid (bx + cy)$ for any $x, y \in \mathbb{Z}$.*

Proof. $\exists k_1 \in \mathbb{Z}$, such that $\frac{b}{a} = k_1$, and

$\exists k_2 \in \mathbb{Z}$, such that $\frac{c}{a} = k_2$.

$\forall x, y \in \mathbb{Z}$:

$$\begin{aligned} xk_1 + yk_2 &\in \mathbb{Z} \\ x\left(\frac{b}{a}\right) + y\left(\frac{c}{a}\right) &= \frac{bx + cy}{a} \in \mathbb{Z} \end{aligned}$$

So $a \mid (bx + cy)$ for any $x, y \in \mathbb{Z}$. □

Proof portfolio part II

Lemma 6. *$4^n - 1$ is divisible by 3 for every natural number n .*

Proof. (By Induction)

1. Base cases:

$$4^1 - 1 = 3, 3 \mid 3.$$

2. Induction Steps: Let $n \in \mathbb{N}$, $4^n - 1$ is divisible by 3.

$$\Rightarrow 4^n - 1 \equiv 0 \pmod{3}$$

$$\Rightarrow 4^n \equiv 1 \pmod{3}$$

$$4^{n+1} - 1 \equiv 4(1) - 1 \equiv 0 \pmod{3}$$

$$\Rightarrow 4^{n+1} - 1 \text{ is divisible by 3.}$$

$4^n - 1$ is divisible by 3, $\forall n \in \mathbb{N}$. □

Definition. Let $A = \{(x, y) \in \mathbb{R}^2 : 2 - x - 2y < (2 - x)(1 - y)\}$.

$$2 - x - 2y < 2 - x - 2y + xy \quad 0 < xy \Rightarrow xy > 0$$

So x, y have the same sign.

Lemma 7.

$$(A \cup B) - C \subseteq (A - (B \cup C)) \cup (B - (A \cap C))$$

Proof. Let $x \in$ Left Hand Side. $x \in (A \cup B) - C \iff ((x \in A) \vee (x \in B)) \wedge (x \notin C)$

if $x \in A$:

$x \notin B$, by $(x \in A) \vee (x \in B)$. Since $x \notin C$, $x \notin (B \cup C)$

$\Rightarrow x \in A - (B \cup C)$

if $x \in B$:

$x \notin A$.

$\Rightarrow x \in B - (A \cap C)$

So $x \in (A - (B \cup C)) \vee x \in (B - (A \cap C))$

$\Rightarrow x \in (A - (B \cup C)) \cup (B - (A \cap C))$

So, $(A \cup B) - C \subseteq (A - (B \cup C)) \cup (B - (A \cap C))$.

□

Lemma 8. Let $f : A \rightarrow B$ be a function and $X, Y \subseteq A$. Then $f(X \cap Y) \subseteq f(X) \cap f(Y)$ with equality when f is injective.

Proof. Let $b \in f(X \cap Y)$,

then $\exists a \in X \cap Y$, such that $f(a) = b$

$\Rightarrow (a \in X) \wedge (a \in Y)$

$b = f(a) \in f(X)$ and $b = f(a) \in f(Y)$

$b \in f(X) \cap f(Y)$

$\Rightarrow f(X \cap Y) \subseteq f(X) \cap f(Y)$

If f is injective:

Let $d \in f(X) \cap f(Y)$

$\exists c_1 \in X, c_2 \in Y, f(c_1) = f(c_2) = d$

Since f is injective, $c_1 = c_2$.

$\Rightarrow c_1 \in X$ and $c_1 = c_2 \in Y$

So $c_1 \in X \cap Y$

Where $d = f(c_1) \in f(X \cap Y)$

So $f(X) \cap f(Y) \subseteq f(X \cap Y)$.

$f(X) \cap f(Y) = f(X \cap Y)$

□

Lemma 9. Suppose that $q \neq 1$ is a real number. Then

$$1 + q + q^2 + \cdots + q^n = \frac{1 - q^{n+1}}{1 - q} \quad \text{for all } n \in \mathbb{Z}, n \geq 0$$

Proof. (Induction)

Base cases:

$$1 + q^0 = \frac{1 - q^1}{1 - q}$$

$1 = 1$, true.

Induction steps:

Assume:

$$1 + q + q^2 + \cdots + q^n = \frac{1 - q^{n+1}}{1 - q}$$

$$1 + q + q^2 + \cdots + q^n + q^{n+1} = \frac{1 - q^{n+1}}{1 - q} + q^{n+1}$$

$$1 + q + q^2 + \cdots + q^n + q^{n+1} = \frac{1 - q^{n+1}}{1 - q} + \frac{q^{n+1}(1 - q)}{1 - q}$$

$$1 + q + q^2 + \cdots + q^n + q^{n+1} = \frac{1 - q^{n+1} + q^{n+1} - q^{n+2}}{1 - q}$$

$$1 + q + q^2 + \cdots + q^n + q^{n+1} = \frac{1 - q^{n+2}}{1 - q}$$

So

$$1 + q + q^2 + \cdots + q^n = \frac{1 - q^{n+1}}{1 - q} \quad \text{for all } n \in \mathbb{Z}, n \geq 0$$

□

Lemma 10. *Let f be a function such that $f(xy) = xf(y) + yf(x)$ for all $x, y \in \mathbb{R}$. Then $f(1) = 0$ and $f(u^n) = nu^{n-1}f(u)$ for all $u \in \mathbb{R}$ and $n \in \mathbb{N}$.*

Proof.

$$f(1) = f(1 \cdot 1) = 1f(1) + 1f(1) = 2f(1)$$

$$2f(1) - f(1) = f(1) = 0$$

(Induction)

Base cases:

$$f(u^1) = f(u)$$

$$1u^0 f(u) = f(u)$$

Equality holds.

Induction steps:

Assume $f(u^n) = nu^{n-1}f(u)$.

$$f(u^{n+1}) = f(u \cdot u^n) = uf(u^n) + u^n f(u) = u \cdot nu^{n-1}f(u) + u^n f(u)$$

$$= (nu^n + u^n)f(u) = (n+1)u^n f(u)$$

So, $f(u^n) = nu^{n-1}f(u)$ for all $u \in \mathbb{R}$ and $n \in \mathbb{N}$.

□

Lemma 11. *Let P be the set of polynomials with real coefficients and with degree ≥ 1 . If $f \in P$, let $A_f := \{x \in \mathbb{R} : f(x) = 0\}$.*

Then $\bigcup_{f \in P} A_f = \mathbb{R}$ and $\bigcap_{f \in P} A_f = \emptyset$.

Proof. I. $\bigcup_{f \in P} A_f = \mathbb{R}$

i. $\mathbb{R} \subseteq \bigcup_{f \in P} A_f.$

$\forall a \in \mathbb{R}, f_a(x) := x - a \in P.$

$$f_a(a) = a - a = 0$$

So, $\forall a \in \mathbb{R}, \exists f_a \in P, f_a(a) = 0.$

$$\Rightarrow \forall a \in \mathbb{R}, a \in \bigcup_{f \in P} A_f$$

$$\mathbb{R} \subseteq \bigcup_{f \in P} A_f.$$

ii. $\bigcup_{f \in P} A_f \subseteq \mathbb{R}.$

Assume $b \in \bigcup_{f \in P} A_f.$

$$\forall f \in P, A_f \subseteq \mathbb{R}$$

$b \in A_f$, for some $f \in P$, so $b \in \mathbb{R}.$

$$\bigcup_{f \in P} A_f \subseteq \mathbb{R}$$

By i. and ii., $\bigcup_{f \in P} A_f = \mathbb{R}.$

II. $\bigcap_{f \in P} A_f = \emptyset.$

(Contradiction)

Assume $\exists c \in \mathbb{R}, \forall f \in P, f(c) = 0.$

Let $f_1(x) := x - 1 \in P, f_2(x) := x - 2 \in P.$

Since $f_1(c) = 0, c - 1 = 0, c = 1.$

Since $f_2(c) = 0, c - 2 = 0, c = 2.$

Contradiction, there is no such c in $\bigcap_{f \in P} A_f.$

$$\Longleftrightarrow \bigcap_{f \in P} A_f = \emptyset$$

□

Lemma 12. *The relation $\sim$ defined on $\mathbb{Z} \times \mathbb{N}$ by*

$$(a, b) \sim (c, d) \Longleftrightarrow ad = bc$$

is an equivalence relation.

Proof. i. Reflexive

Let $a \in \mathbb{Z}$, and $b \in \mathbb{N}$.

$$(a, b) \sim (a, b) \iff ab = ba$$

And $ab = ba$ by properties of multiplication under $\mathbb{R}$, and $\mathbb{Z}, \mathbb{N} \subset \mathbb{R}$, so $\sim$ is reflexive

ii. Symmetric

Let $c \in \mathbb{Z}$, and $d \in \mathbb{N}$, assume

$$(a, b) \sim (c, d)$$

Where $ad = bc$.

Since $ad = bc \iff da = cb$,

$$da = cb \iff (c, d) \sim (a, b)$$

So $\sim$ is symmetric.

iii. Transitive

Let $e \in \mathbb{Z}$, and $f \in \mathbb{N}$, assume $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$.

$\Rightarrow ad = bc, cf = de$.

Since $b \in \mathbb{N}$, $b \neq 0$.

$c = \frac{ad}{b}$, and substitute it in second equation.

$$\frac{ad}{b}f = de$$

$$adf = db e$$

Since $d \in \mathbb{N}$, $d \neq 0$.

$$af = de$$

$$(a, d) \sim (e, f)$$

$\sim$ is transitive.

So $\sim$ is an equivalence relation.

□